

Notes: Maksimovic and Phillips (JF, 1998)

Asset Efficiency and Reallocation Decisions of Bankrupt Firms

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Economic framework	3
3.1	Why industry conditions matter	3
3.2	Potential indirect costs of Chapter 11	3
4	Data and measurement	3
4.1	Bankruptcy sample and Census plant data	3
4.2	Productivity and cash-flow measures	4
4.3	Industry conditions	4
5	Empirical strategy	4
5.1	Productivity around Chapter 11	4
5.2	Composition correction	4
5.3	Asset sale and closure regressions	5
6	Main results	5
7	Original tables with reading notes	6
7.1	Table I – Chapter 11 Bankruptcy Sample Characteristics; Table II – Industry-Adjusted Productivity of Plants of Chapter 11 Firms	6
7.2	Table III – Plant-level Operating Cash Flows of Chapter 11 Firms; Table IV – Productivity Controlling for Composition Changes	7
7.3	Table V – Asset Sales by Chapter 11 Firms, Parts 1 and 2	8
7.4	Table V – Asset Sales by Chapter 11 Firms, Part 3; Table VI – Asset Sale Decisions, Part 1	9
7.5	Table VI – Asset Sale Decisions, Part 2; Table VII – Plant Asset Sales: Estimated Probabilities	10
7.6	Table VIII – Plant Closing Decisions, Parts 1 and 2	11
7.7	Table IX – Plant Closures: Estimated Probabilities; Table X – Transfers to Chapter 7 Liquidation, Part 1	12
7.8	Table X – Transfers to Chapter 7 Liquidation, Part 2	13
8	How to read the paper’s causal claims	13
9	Key lessons for future research	14
10	Short summary	14

1 Big picture

- **Question.** Does Chapter 11 bankruptcy create large real economic costs, or does it mainly allow firms to reorganize and redeploy assets according to ordinary economic fundamentals?
- **Core idea.** Bankruptcy outcomes cannot be evaluated without industry conditions. The opportunity cost of keeping, selling, or closing a plant depends on industry demand, capacity utilization, R&D intensity, and plant productivity.
- **Main finding.** Chapter 11 status by itself explains surprisingly little of plant productivity, asset sales, and plant closures once industry conditions and plant-level productivity are considered. Industry conditions are much more important.
- **Key distinction.** In declining industries, Chapter 11 firms are common and bankruptcy appears to facilitate capacity exit; plants of bankrupt firms are not systematically less productive than industry peers. In high-growth industries, bankrupt plants are less productive, and sold plants often become more productive under new owners.
- **Bottom line.** The paper finds little evidence of large indirect bankruptcy costs for firms that enter Chapter 11. However, the selection of which firms enter bankruptcy differs sharply across industry states, so the absence of realized costs in Chapter 11 does not imply that all distressed firms would face low costs if they filed.

2 Incremental contribution of the paper

- **Plant-level view of bankruptcy.** Earlier studies often use firm-level outcomes and focus on firms that survive or emerge. This paper uses plant-level Census data, allowing the authors to track individual plants as firms sell, close, retain, or transfer them.
- **No survivorship-only inference.** The sample includes firms that do not emerge from Chapter 11 and firms that convert to Chapter 7. This is important because focusing only on emerged firms would mechanically drop bad outcomes.
- **Industry equilibrium logic.** The paper treats asset reallocation as an industry-level problem. The same plant has a different opportunity cost in a growing industry than in a shrinking industry.
- **Separates two potential bankruptcy costs.** The paper distinguishes (i) deterioration of plant efficiency while in Chapter 11 and (ii) distorted reallocation decisions, such as inefficient delays in asset sales or closures.
- **Tests asset redeployment directly.** Because the data identify plant ownership changes and plant closures, the paper tests whether plants sold by bankrupt firms improve or deteriorate under new owners.
- **Broader governance implication.** The paper argues that assessments of bankruptcy, takeover, or governance mechanisms must condition on industry demand. A mechanism that looks weak in a declining industry may have a different role when demand is high.

3 Economic framework

3.1 Why industry conditions matter

The model logic is built around the marginal product and opportunity cost of capital. Firms differ in their ability to manage plants. More productive firms can produce more output from the same inputs, but managing a larger set of plants involves control costs. Therefore, the value of reallocating a plant depends on both the plant's own productivity and the industry state.

- When demand is high, assets are valuable. Transferring a plant from a low-productivity owner to a high-productivity owner can generate large gains.
- When demand is low, industry-wide excess capacity lowers the value of assets. The gain from transferring plants across owners is smaller, and the efficient action may be closure or scrapping.
- Because cash flows move with industry demand, the firms that enter bankruptcy in high-growth industries are likely to be different from the firms that enter bankruptcy in declining industries.

Economic message: Bankruptcy is not a single uniform treatment. It occurs in different economic environments. A Chapter 11 filing in a declining industry may reflect excess capacity, while a filing in a high-growth industry is more likely to reveal firm-specific inefficiency.

3.2 Potential indirect costs of Chapter 11

The paper tests two broad classes of real bankruptcy costs.

1. **Productivity deterioration.** While under bankruptcy protection, firms may undermaintain assets, lose customers, or fail to make investments. If this is important, the productivity of retained plants should decline relative to industry peers.
2. **Distorted reallocation.** Chapter 11 may change incentives or control rights in ways that delay optimal plant sales or closures. If this is important, bankrupt firms should sell or close plants differently from comparable industry firms after controlling for productivity and industry demand.

Interpretation: The authors' null is not simply that bankrupt firms look like healthy firms. Bankrupt firms are selected. The more precise null is that, after conditioning on plant productivity and industry conditions, Chapter 11 itself does not create large additional distortions.

4 Data and measurement

4.1 Bankruptcy sample and Census plant data

The paper examines manufacturing firms that filed for Chapter 11 between 1978 and 1989. The key data source is the Longitudinal Research Database at the U.S. Census Bureau, which provides plant-level information on output, inputs, employment, ownership, plant closures, and plant sales. The main empirical window is 1977 to 1990.

- **Bankrupt firms:** 302 Chapter 11 firms are matched to Census plant records.
- **Plant observations:** the paper tracks plants of bankrupt firms and more than 50,000 plants of nonbankrupt manufacturing firms.

- **Key advantage:** a plant can be followed after sale, so productivity under old and new owners can be compared.
- **Outcome classification:** firms may emerge from Chapter 11, remain in Chapter 11, or transfer to Chapter 7 liquidation.

4.2 Productivity and cash-flow measures

The main performance measure is total factor productivity (TFP). The authors estimate a translog production function by industry and define a plant's TFP as actual output relative to predicted output, standardized by the industry standard deviation. This avoids relying solely on accounting cash flows and allows comparisons of plant efficiency within industries.

Operating cash flow is measured at the plant level as shipments minus labor, materials, and energy costs, divided by shipments. This is an operating-margin measure rather than full corporate cash flow because overhead and SG&A are not allocated to plants.

4.3 Industry conditions

The paper groups industries into quartiles based on long-run changes in real industry shipments. The first quartile is the declining-industry group, and the fourth quartile is the high-growth group.

- **Industry shipments:** proxy for demand and the value of capacity.
- **Capacity utilization:** proxy for excess capacity and the pressure to close plants.
- **Industry R&D and advertising:** proxy for asset specificity, technological change, and difficulty of valuing businesses.

5 Empirical strategy

5.1 Productivity around Chapter 11

The first set of tests compares bankrupt plants' industry-adjusted productivity before, during, and after Chapter 11. The paper reports these comparisons overall and separately for declining and high-growth industry quartiles.

Identification logic: If Chapter 11 itself damages firms, productivity should fall while firms are under bankruptcy protection, especially for plants that are retained. If low productivity is mostly selection into bankruptcy, productivity should be low before filing but should not necessarily decline afterward.

5.2 Composition correction

A key concern is that the composition of bankrupt firms' assets changes over time because plants can be sold or closed. The paper therefore follows the same set of plants over time depending on how long they remain in Chapter 11. This controls for survivorship and composition bias.

Takeaway: Without plant-level composition corrections, a researcher could mistake asset selection for productivity deterioration. The paper's plant-level design is crucial for separating these forces.

5.3 Asset sale and closure regressions

The paper estimates logistic regressions for whether a plant is sold or closed in a given year. Explanatory variables include industry conditions, plant productivity, plant cash flow, firm size, plant age, and bankruptcy status dummies. The regressions interact bankruptcy status with productivity and industry variables to test whether Chapter 11 changes the sensitivity of sale and closure decisions.

6 Main results

1. **Bankruptcy incidence is highest in declining industries.** The share of industry plants in Chapter 11 falls from 3.23% in the declining quartile to 0.98% in the high-growth quartile.
2. **Declining-industry bankrupt plants do not deteriorate.** In low-growth industries, Chapter 11 plants do not exhibit significant productivity declines relative to industry counterparts.
3. **High-growth bankrupt plants are different.** In high-growth industries, plants of Chapter 11 firms are more likely to be less productive than industry peers, and plants that remain in Chapter 11 for a long time show declines.
4. **Asset sales are tied to growth and productivity.** Plant sales are more valuable and more common in high-growth industries. Sold plants in high-growth industries tend to improve under new owners.
5. **Bankruptcy status has limited marginal power for asset sales.** Once industry and plant characteristics are controlled for, Chapter 11 status does not strongly increase or decrease the probability of plant sale.
6. **Plant closures are concentrated in declining industries.** Chapter 11 is associated with capacity exit in declining industries, but this appears to be a response to low demand rather than evidence that bankruptcy protects inefficient plants.
7. **Chapter 7 transfers are lower-productivity cases.** Firms that later transfer from Chapter 11 to Chapter 7 have much lower productivity, consistent with liquidation selecting genuinely weak assets.

7 Original tables with reading notes

7.1 Table I – Chapter 11 Bankruptcy Sample Characteristics; Table II – Industry-Adjusted Productivity of Plants of Chapter 11 Firms

Table I
Chapter 11 Bankruptcy Sample Characteristics

Sample characteristics of plants of bankrupt firms that declared Chapter 11 in the years 1978 to 1989. We obtain plant-level data from the Annual Survey of Manufacturers (ASM), Bureau of the Census, U.S. Department of Commerce. Total factor productivity (TFP) statistics are given for the year prior to the declaration of Chapter 11 for each of the bankrupt firms. TFP is a relative measure of productivity calculated such that average TFP over time equals 0. TFPs are standardized by dividing each TFP by the standard deviation of the industry's TFP at the four-digit level. Shipment or output change is determined by using aggregate industry data from the NBER for manufacturing industries. We determine quartiles by computing the change in the real value of shipments over 10 years at the three-digit SIC code level. We use the year of the first bankruptcy in an industry when there are multiple bankruptcies in an industry for the number of nonbankrupt industry firms and industry plants. Average plant age is calculated as the year before Chapter 11 less the time the plant first appeared in the database, beginning with 1972. (Standard errors of the means are in parentheses.)

	Sample of Firms	
	Bankrupt Firms	Nonbankrupt Industry Firms
Panel A: Characteristics of Plants of Bankrupt and Nonbankrupt Firms		
Number of plants (302 bankrupt firms)	1,075	56,256
Average plant size (Value of shipments in \$ millions)	61.954	29.764
Average plant age (years)	11.5 (0.154)	8.24 (0.025)
Total factor productivity: Year before Chapter 11		
Avg. standardized TFP (bankrupt - industry avg.)	-0.048* (0.028)	
Panel B: Industry Characteristics: Growth in Industry Shipments Quartiles		
Quartile 1: Number of plants	259	8,021
% of industry plants in bankruptcy	3.23%	
Median growth in real industry shipments	-33.93%	
Quartile 2: Number of plants	457	18,089
% of industry plants in bankruptcy	2.53%	
Median growth in real industry shipments	-4.12%	
Quartile 3: Number of plants	199	12,978
% of industry plants in bankruptcy	1.53%	
Median growth in real industry shipments	8.11%	
Quartile 4: Number of plants	160	16,295
% of industry plants in bankruptcy	0.98%	
Median growth in real industry shipments	23.14%	

* Significantly different from zero at the 10 percent level using a two-tailed test for the difference of the mean from zero.

Table I – Chapter 11 Bankruptcy Sample Characteristics

Table II
Industry-Adjusted Productivity of Plants of Chapter 11 Firms

This table presents summary statistics that examine whether the productivity and number of plants of bankrupt firms varies over aggregate industry quartiles. Quartiles are defined by change in industry demand over the 10-year period, using the change in three-year averages for 1977–1979 to 1986–1988. Numbers are average relative total factor productivity (TFP) less the industry average TFP for firms that declared Chapter 11 in the years 1978 to 1989. We standardize TFPs by dividing each TFP by the standard deviation of the industry TFPs at the four-digit code level. We determine quartiles by using aggregate industry shipment data from the NBER for manufacturing industries. The industry shipment data is the real value of shipments at the three-digit SIC code level. Q1 is productivity for quartile 1, Q4 is productivity for quartile 4. (*t*-statistics for significant differences from zero are in parentheses.)

	Average Productivity		Average Productivity by Industry Growth Quartile		<i>t</i> -statistics for Significant Difference for Q4 – Q1	
	All Firms	<i>n</i>	Q1 (lowest quartile)	<i>n</i>	Q4	<i>n</i>
Panel A: Years Before Chapter 11						
Year -2	-0.044 (-1.53)	1188	-0.052 (-1.04)	308	-0.009 (-0.12)	196 (0.47)
Year -1	-0.048* (-1.70)	1075	-0.032 (-0.53)	259	-0.151* (-1.70)	160 (-1.14)
Panel B: In Chapter 11						
Year 0	-0.092* (-2.51)	873	-0.118* (-1.73)	207	-0.188 (-1.46)	154 (-0.56)
<i>t</i> -stat for average	(-0.89)		(-0.98)		(-0.24)	
Year +1	0.036 (0.86)	639	0.086 (1.06)	156	0.015 (0.13)	102 (-1.26)
<i>t</i> -stat for average	(1.67)		(1.20)		(1.14)	
Year +2	-0.024 (-0.51)	345	0.072 (0.72)	99	-0.237* (-1.74)	61 (-1.98) ^b
<i>t</i> -stat for average	(0.42)		(1.04)		(-0.53)	
Year +3	-0.071 (-1.14)	200	0.027 (0.31)	72	-0.369* (-2.47)	29 (-2.62) ^a
<i>t</i> -stat for average	(-0.33)		(0.57)		(-1.26)	
Year +4	-0.023 (-0.31)	143	0.093 (0.87)	56	-0.335* (-2.66)	24 (-2.40) ^a
<i>t</i> -stat for average	(0.33)		(1.03)		(-1.19)	
Panel C: After Emerging from Chapter 11						
Year +1	-0.123* (-1.89)	372	-0.055* (-0.48)	75	0.022 (0.17)	49 (0.44)
<i>t</i> -stat for average	(-0.83)		(-0.18)		(1.13)	
Year +2	-0.093 (-1.48)	213	-0.040 (-0.31)	56	-0.003 (0.02)	34 (0.65)
<i>t</i> -stat for average	(-0.65)		(-0.06)		(0.90)	
Year +3	-0.165* (-1.63)	124	-0.003 (-0.01)	28	-0.288 (-1.47)	30 (-0.63)
<i>t</i> -stat for average	(-1.11)		(0.10)		(-0.64)	

^{a,b,c} Significantly different from zero at the 1 percent, 5 percent, and 10 percent levels, respectively, using a two-tailed *t*-test.

Table II – Industry-Adjusted Productivity of Plants of Chapter 11 Firms

Read: Table I defines the empirical sample and the industry-growth quartiles. Bankrupt-firm plants are larger and older than nonbankrupt industry plants. The frequency of Chapter 11 is much higher in declining industries: 3.23% of industry plants are in bankruptcy in quartile 1, compared with 0.98% in quartile 4.

Interpretation: Table II shows that average industry-adjusted productivity is low for Chapter 11 plants in the year before filing, but this masks sharp heterogeneity. In the declining quartile, productivity is not significantly below the industry average. In the high-growth quartile, productivity is lower and becomes especially weak for longer Chapter 11 spells.

Takeaway: The raw fact that bankrupt firms have lower productivity is not enough. The key question is where and when they file. Bankruptcy in a declining industry has a different meaning from bankruptcy in a high-growth industry.

7.2 Table III – Plant-level Operating Cash Flows of Chapter 11 Firms; Table IV – Productivity Controlling for Composition Changes

Table III
Plant-level Operating Cash Flows of Chapter 11 Firms

This table presents the cash flow of plants of bankrupt firms by aggregate industry quartiles. Quartiles are defined by the real value industry shipments over a ten-year period. Cash flow numbers are plant-level values of shipments less materials, energy, and labor costs. We present changes relative to the year prior to bankruptcy, as all averages (levels) are significantly different from zero at the 1 percent level. We determine quartiles by using aggregate industry shipment data from the NBER for manufacturing industries. The change in shipments is calculated over a ten-year period at the three-digit SIC code level, using the change in three-year averages for 1977–1979 to 1986–1988. Q1 is productivity for quartile 1, Q4 is productivity for quartile 4. (*t*-statistics for significant differences from zero are in parentheses.)

	Cash Flows by Industry Growth Quartile					<i>t</i> -statistics for Significant Difference for Q4 – Q1
	All Firms	<i>n</i>	Q1 (lowest quartile)	<i>n</i>	Q4	<i>n</i>
	Panel A: Years Before Chapter 11—Average Cash Flow					
Year –2	0.190 ^a (19.19)	1188	0.167 ^a (10.17)	308	0.233 ^a (13.73)	196 (2.81) ^a
Year –1	0.203 ^a (24.38)	1075	0.144 ^a (7.36)	259	0.212 ^a (13.90)	160 (2.75) ^a
Panel B: In Chapter 11—Change in Cash Flow from Year before Chapter 11						
Change Year –1 to 0	–0.007 (–0.48)	607	0.023 (0.81)	207	–0.002 (–0.08)	13 (–0.68)
Year –1 to +1	0.039 ^a (2.96)	639	0.080 ^a (2.82)	156	0.027 (1.21)	102 (–1.85) ^c
Year –1 to +2	0.021 (1.42)	345	0.067 ^a (2.40)	99	–0.020 (–0.75)	61 (–2.12) ^b
Year –1 to +3	–0.016 (–0.90)	200	0.049 ^c (1.68)	72	–0.043 ^c (–1.64)	29 (–1.69) ^c
Year –1 to +4	0.014 (0.69)	143	0.051 (1.57)	56	–0.016 (–0.75)	24 (–1.48)
Panel C: After Emerging from Chapter 11						
Year –1 to +1	0.013 (1.29)	372	0.056 (1.32)	75	0.043 (1.27)	49 (–0.26)
Year –1 to +2	–0.010 (–0.41)	213	0.101 ^b (1.96)	56	–0.024 (–0.51)	34 (–1.93) ^c
Year –1 to +3	–0.054 (–1.29)	124	0.034 (0.31)	28	–0.017 (–0.34)	30 (–0.43)

^{a,b,c} Significantly different from zero at the 1 percent, 5 percent, and 10 percent levels, respectively, using a two-tailed *t*-test.

Table III – Plant-level Operating Cash Flows of Chapter 11 Firms

Table IV
Productivity Controlling for Composition Changes

This table presents the prior productivity of plants by the number of years in Chapter 11, for industry growth quartiles 1 and 4. The table thus directly controls for changing asset composition. Plants leave Chapter 11 because they are sold off or closed down, or because the firm emerges from Chapter 11. Numbers are plant-level total factor productivity (TFP) less the industry average TFP for firms that declared Chapter 11 in the years 1978 to 1989. TFPs are standardized by dividing each TFP by the standard deviation of the industry TFPs at the four-digit SIC code level. We determine quartiles by using the change in ten-year shipments at the aggregate three-digit SIC code level for all industries. Plant counts do not add up exactly to the yearly changes in plants in Table II because of missing plant-year data. Q1 represents productivity of plants in quartile 1, Q4 for quartile 4, respectively. Q4 – Q1 represents the difference in means between quartile 4 and quartile 1. (*t*-statistics for significant differences from 0 are in parentheses.)

Number of Years in Chapter 11	Average Year -1	Change				
		Year -1 to 0	-1 to 1	-1 to 2	-1 to 3	-1 to 4
Panel A: Industry Quartile 1: Productivity in Bankruptcy						
< 1 year, <i>n</i> = 47	-0.293 ^b (-2.14)	0.068 (0.31)				
<i>t</i> -stat for Q1						
1 year, <i>n</i> = 54	-0.287 ^b (-2.19)	0.187 (0.96)	0.312 (1.48)			
<i>t</i> -stat for Q1						
2 years, <i>n</i> = 27	-0.086 (-0.67)	-0.110 (-0.55)	0.252 (1.18)	0.240 (0.23)		
<i>t</i> -stat for Q1						
3 years, <i>n</i> = 13	0.286 (1.44)	-0.266 (-0.86)	-0.172 (-0.63)	-0.162 (-0.53)	0.001 (0.01)	
<i>t</i> -stat for Q1						
4 or more years, <i>n</i> = 56	-0.078 (-0.71)	-0.023 (-0.21)	0.148 (1.18)	0.087 (0.88)	0.107 (1.02)	0.176 (1.55)
<i>t</i> -stat for Q1						
Panel B: Industry Quartile 4: Productivity in Bankruptcy						
< 1 year, <i>n</i> = 20	-0.306 (-1.61)	-0.517 (-1.71) ^c				
<i>t</i> -stat for Q4						
<i>t</i> -stat for Q4 - Q1						
1 year, <i>n</i> = 39	-0.159 (-0.68)	0.233 (0.66)	0.212 (0.68)			
<i>t</i> -stat for Q4						
<i>t</i> -stat for Q4 - Q1						
2 years, <i>n</i> = 29	0.247 (0.99)	0.057 (0.18)	0.055 (0.17)	-0.352 (-1.02)		
<i>t</i> -stat for Q4						
<i>t</i> -stat for Q4 - Q1						
3 years, <i>n</i> = 6	-0.541 (-1.51)	-0.239 (-0.60)	-0.251 (-0.64)	0.029 (0.07)	0.128 (0.34)	
<i>t</i> -stat for Q4						
<i>t</i> -stat for Q4 - Q1						

Table IV – Productivity Controlling for Composition Changes

Read: Table III shows that cash flow levels are positive and often higher in high-growth industries even when productivity is weak. For example, a low-productivity plant in a high-growth industry can still have high cash flows because the industry is strong.

Interpretation: Table IV is one of the paper's most important diagnostics. It follows plants by the number of years they remain in Chapter 11, controlling for asset sales and closures. In quartile 1, there is no evidence of significant productivity deterioration during Chapter 11. In quartile 4, productivity falls for plants that remain in Chapter 11 for four or more years.

Takeaway: Composition matters. Productivity changes in bankrupt firms can be explained largely by which plants are sold, closed, or retained. Chapter 11 does not generally appear to destroy retained-plant efficiency, except for long stays in high-growth industries.

7.3 Table V – Asset Sales by Chapter 11 Firms, Parts 1 and 2

Table V
Asset Sales by Chapter 11 Firms

This table examines the productivity of firms' plants that were sold by Chapter 11 firms and industry plants in the same four-digit SIC code. I use external industry data from the NBER and represent the change in shipments over a ten-year period at the three-digit SIC code level. Total factor productivity (TFP) is standardized by dividing each TFP by the standard deviation of the industry TFPs at the four-digit SIC code level. Years after emergence are combined for the lowest two and highest two quartiles because of disclosure restrictions. Industry numbers do not include takeovers or mergers. Asset sale firms are defined as firms selling assets while remaining in business (t-statistics are in parentheses.)

Panel A: Before Chapter 11								
	Lowest Two Quartiles				Highest Two Quartiles			
	Bankrupt Firms		Industry Firms		Bankrupt Firms		Industry Firms	
	Old Owners	New Owners	Old Owners	New Owners	Old Owners	New Owners	Old Owners	New Owners
Asset sale firms: TFP of sales	-0.014	-0.027	-0.015	0.017	-0.065	-0.052	-0.032	-0.0
t-stat for average TFP = 0	(-0.36)	(-0.46)	(-0.98)	(1.23)	(-0.91)	(-0.60)	(-2.07) ^b	(-1.1)
t-stat for improvement under new owners, relative to industry	(-2.18) ^b				(0.03)			
Number of plants [percent sold off] [percent based on total plant years]	193 [41.3%] [11.1%]		706 [18.4%] [6.2%]		122 [52.4%] [14.9%]		502 [18.1%] [5.8%]	
Assets retained: TFP	-0.049		0.029		-0.057		0.040	
t-stat for average TFP = 0	(-1.26)		(1.50)		(-1.25)		(2.28) ^b	
t-stat for retained-sold	(-0.54)		(1.78) ^c		(0.09)		(3.08) ^a	
Number of plants	274		3124		111		3324	
Nonasset sale firms: TFP	-0.070		-0.008		-0.016		-0.009	
t-stat for average TFP = 0	(-1.63)		(-1.15)		(-0.57)		(-1.23)	
Number of plants	434		19664		251		16967	

Table V – Asset Sales by Chapter 11 Firms, Part 1

Panel B: Chapter 11 Period								
	Lowest Two Quartiles				Highest Two Quartiles			
	Bankrupt Firms		Industry Firms		Bankrupt Firms		Industry Firms	
	Old Owners	New Owners	Old Owners	New Owners	Old Owners	New Owners	Old Owners	New Owners
Asset sale firms: TFP of sales	-0.030	-0.057	-0.033	-0.008	0.011	0.080	-0.009	-0.0
t-stat for average TFP = 0	(-0.36)	(-1.45)	(-2.24) ^b	(-0.44)	(0.11)	(1.72) ^c	(-0.65)	(-3.3)
t-stat for improvement under new owners, relative to industry	(-1.20)				(2.29) ^b			
Number of plants [percent sold off] [percent based on total plant years]	138 [44.4%] [16.5%]		437 [20.2%] [8.8%]		47 [58.8%] [24.4%]		466 [19%] [9.9%]	
Assets retained: TFP	0.053		0.046		-0.209		0.035	
t-stat for average TFP = 0	(1.56)		(1.53)		(-2.37) ^b		(2.66) ^a	
t-stat for retained-sold	(0.93)		(2.38) ^a		(-1.66) ^c		(2.34) ^b	
Number of plants	172		1726		33		1983	
Nonasset sale firms: TFP	0.064		-0.002		-0.154		-0.005	
t-stat for average TFP = 0	(1.19)		(-0.26)		(-1.38)		(-0.68)	
Number of plants	558		21592		120		19738	

Panel C: After Chapter 11								
	Lowest Two Quartiles				Highest Two Quartiles			
	Bankrupt Firms	Number of Plants	Industry Firms	Number of Plants	Bankrupt Firms	Number of Plants	Industry Firms	Number of Plants
Productivity of Assets Retained	0.006	167	0.027	1472	-0.167	85	0.020	180
t-stat for average TFP = 0	(0.15)		(2.50) ^a		(-1.72) ^c		(2.50) ^a	

Table V – Asset Sales by Chapter 11 Firms, Part 2

Read: Table V compares sold and retained plants for Chapter 11 firms and industry firms. The table separates periods before, during, and after Chapter 11, and splits low-growth and high-growth industries.

Interpretation: The central message is that asset sales are more important in high-growth industries. In high-growth industries, plants sold by bankrupt firms improve under new owners. Retained plants in those industries are relatively weaker. This points to value-enhancing reallocation rather than indiscriminate fire sales.

Takeaway: Chapter 11 does not appear to block useful asset transfers. When the value of redeployment is high, the data show that sold assets can become more productive under new owners.

7.4 Table V – Asset Sales by Chapter 11 Firms, Part 3; Table VI – Asset Sale Decisions, Part 1

Table V—Continued

Panel D: Productivity of Purchasers' Existing Assets (for the four years surrounding purchase)				
	Lowest Two Quartiles		Highest Two Quartiles	
	Quartiles 1 and 2	Number of Plants	Quartiles 3 and 4	Number of Plants
Productivity of Purchaser				
TFP before purchase	0.069	1396	0.055	1554
t-stat for average TFP = 0	(2.83) ^a		(2.74) ^a	
TFP after purchase	0.085	1274	0.030	1405
t-stat for average TFP = 0	(2.07) ^b		(1.18)	

^{a,b,c} Significantly different from 0 at the 1 percent, 5 percent, and 10 percent levels, respectively, using a two-tailed t-test for the difference in mean from zero.

Table V – Asset Sales by Chapter 11 Firms, Part 3

Table VI
Asset Sale Decisions

Regressions test the effects of plant-level productivity and industry-level demand and supply variables on asset sale decisions of bankrupt firms and other nonbankrupt industry firms. We estimate the regressions using a logistic-limited dependent variable model. The dependent variable equals 1 if a plant is sold in that year. The change in industry shipments and capacity utilization are yearly at the four-digit SIC code level. Weighted industry R&D is from COMPUSTAT firm-level data and represents R&D and advertising expense divided by firm sales aggregated up to the three-digit SIC code. We calculate total factor productivity (TFP) using a translog production function. Operating cash flow is the value of shipments less labor, materials, and energy costs divided by value of shipments. Plant scale is the plant's asset size divided by the average assets for plants in each industry. Data are yearly from 1977 to 1990. (*p*-values are in parentheses.)

	Dependent Variable: Plant Asset Sale			Test for Significant Difference: Quartile 4 – Quartile 1 (<i>p</i> -value)
	Logit A: Full Sample	Logit B: Quartile 1	Logit C: Quartile 4	
Industry-level variables				
Change in industry shipments	1.141 (0.000) ^a	0.983 (0.006) ^a	0.538 (0.161)	(0.000) ^a
Weighted industry R&D	1.935 (0.000) ^a	9.675 (0.000) ^a	0.256 (0.773)	(0.000) ^a
Capacity utilization	−0.015 (0.000) ^a	−0.010 (0.000) ^a	−0.005 (0.129)	(0.135)
Standard deviation of industry shipments	−4.744 (0.000) ^a	1.381 (0.170)	−3.489 (0.001) ^a	(0.002) ^a
Plant- and firm-level variables				
Number of plants owned by firm	−0.005 (0.000) ^a	−0.009 (0.000) ^a	0.000 (0.839)	(0.000) ^a
Log of firm shipments	0.382 (0.000) ^a	0.404 (0.000) ^a	0.378 (0.000) ^a	(0.399)
Total factor productivity (TFP) (Lagged one year)	−0.168 (0.000) ^a	−0.130 (0.005) ^a	−0.132 (0.000) ^a	(0.207)
Plant-level operating cash flow (Lagged one year)	0.173 (0.007) ^a	0.017 (0.901)	0.339 (0.010) ^a	(0.115)
Relative plant scale	−0.154 (0.000) ^a	−0.167 (0.000) ^a	−0.171 (0.000) ^a	(0.905)
Plant age	0.020 (0.000) ^a	0.037 (0.000) ^a	0.019 (0.000) ^a	(0.006)
Bankruptcy variables				
Before Chapter 11 dummy variable (=1 for years before firm is in Ch. 11)	0.602 (0.000) ^a	0.275 (0.073) ^c	0.266 (0.209)	(0.751)
In Chapter 11 dummy variable (=1 while firm is in Ch. 11)	0.409 (0.198)	−0.068 (0.963)	0.654 (0.321)	(0.332)
After Chapter 11 dummy variable (=1 after firm emerges from Ch. 11)	−0.119 (0.583)	−0.950 (0.184)	−0.465 (0.474)	(0.693)
In Chapter 11*TFP	0.251 (0.135)	0.483 (0.214)	0.680 (0.082) ^c	(0.900)
In Chapter 11*cash flow	−0.737	−1.368	−3.209	

Table VI – Asset Sale Decisions, Part 1

Read: The final panel of Table V shows that purchasers' existing plants are above average productivity, both before and after purchase. This means buyers are generally efficient firms rather than weak firms absorbing assets.

Interpretation: Table VI begins the logistic analysis of plant asset sales. Industry shipment growth, industry R&D, productivity, plant scale, age, and cash flow all help explain whether a plant is sold. The pre-Chapter 11 dummy is positive, but the in-Chapter 11 and post-Chapter 11 dummies are not economically dominant.

Takeaway: Asset sales are governed mainly by industry demand and plant characteristics. Bankruptcy status is less important than the economic environment in explaining the sale decision.

7.5 Table VI – Asset Sale Decisions, Part 2; Table VII – Plant Asset Sales: Estimated Probabilities

Table VI—Continued

	Dependent Variable: Plant Asset Sale			Test for Significant Difference: Quartile 4 – Quartile 1 (<i>p</i> -value)
	Logit A: Full Sample	Logit B: Quartile 1	Logit C: Quartile 4	
Bankruptcy variables				
In Chapter 11*industry R&D	0.757 (0.845)	22.800 (0.080) ^c	–14.718 (0.088) ^c	(0.010) ^a
In Chapter 11*change in shipments	0.467 (0.801)	–2.786 (0.358)	5.259 (0.012) ^b	(0.014) ^b
In Chapter 11*capacity utilization	–0.004 (0.315)	–0.010 (0.581)	0.009 (0.369)	(0.365)
Total plant years	371,373	60,477	85,148	
χ^2 statistic	4370.57	653.20	1255.57	
Significance level (<i>p</i> -value)	<1%	<1%	<1%	

^{a,b,c} Significantly different from zero at the 1 percent, 5 percent, and 10 percent levels, respectively.

Table VI – Asset Sale Decisions, Part 2

Table VII

Plant Asset Sales: Estimated Probabilities

Estimated probabilities of an asset sale by bankrupt and nonbankrupt firms at the 10th, 25th, 50th, and 90th percentiles of total factor productivity (TFP). We compute estimated probabilities using the coefficients from the logit regressions from Table VI predicting asset sales. Panel A uses the coefficients from logit regression A from Table VI and the data for the full sample. Panel B is for quartiles 1 and 4 and uses the coefficients from logit regressions B and C from Table VI. We compute these probabilities holding all other variables besides TFP and TFP interaction terms at the sample means of the nonbankrupt and bankrupt firms. For the year $t - 1$, nonbankrupt firms and the bankrupt firms probabilities are computed with the before-bankruptcy indicator variable equal to 0 and 1 respectively. For the year $t + 1$ for the bankrupt firm, we include the during-bankruptcy indicator variable and the interactions with the industry and productivity variables. For quartiles, the percentiles refer to that specific subsample. Thus the 50th percentile for quartile 1 is the 12.5th percentile of the overall sample.

		Probability of Plant Asset Sale at TFP:			
		10th Percentile	25th Percentile	50th Percentile	90th Percentile
Panel A: Full Sample					
Bankrupt firm:	Year $t - 1$	7.65%	7.06%	6.57%	5.59%
(data from bankrupt sample)	Year $t + 1$	4.91%	5.16%	5.39%	5.88%
Bankrupt firm:	Year $t - 1$	3.15%	2.91%	2.70%	2.25%
(data from nonbankrupt sample)	Year $t + 1$	2.16%	2.25%	2.33%	2.54%
Nonbankrupt firm:	Year $t - 1$	1.75%	1.62%	1.50%	1.25%
(data from nonbankrupt sample)	Year $t + 1$	1.85%	1.70%	1.57%	1.31%
Nonbankrupt firm:	Year $t - 1$	4.34%	3.99%	3.71%	3.14%
(data from bankrupt sample)	Year $t + 1$	4.46%	4.02%	3.68%	3.06%
Panel B: Quartiles 1 and 4					
Quartile 1: Bankrupt firm	Year $t - 1$	4.90%	4.62%	4.35%	3.74%
	Year $t + 1$	2.55%	3.23%	3.93%	5.70%
Quartile 1: Nonbankrupt firm	Year $t - 1$	4.48%	4.14%	3.81%	3.14%
	Year $t + 1$	3.64%	3.28%	2.90%	2.22%
Quartile 4: Bankrupt firm	Year $t - 1$	8.09%	7.51%	6.92%	5.74%
	Year $t + 1$	3.74%	4.22%	4.87%	6.64%
Quartile 4: Nonbankrupt firm	Year $t - 1$	4.78%	4.51%	4.25%	3.66%
	Year $t + 1$	4.86%	4.45%	4.15%	3.61%

Table VII – Plant Asset Sales: Estimated Probabilities

Read: The continuation of Table VI reports interactions between Chapter 11 status and industry variables. In high-growth industries, Chapter 11 asset-sale decisions are more sensitive to shipment growth and productivity, while cash-flow interactions suggest that bankrupt high-growth firms may sell efficient plants to raise cash.

Interpretation: Table VII converts coefficients into estimated probabilities. Bankrupt firms have a higher probability of sale, but much of this difference comes from sample means and covariates rather than Chapter 11 status itself. In quartile 4, the probability of asset sale rises with TFP during bankruptcy, consistent with the finding that better plants are sold in high-growth industries.

Takeaway: The economic magnitude reinforces the regression result: bankruptcy status alone is not the central determinant of asset sales. Plant quality and industry state do most of the work.

7.6 Table VIII – Plant Closing Decisions, Parts 1 and 2

Table VIII
Plant Closing Decisions

Regressions test the effects of plant-level productivity and industry-level demand and supply variables on plant closing of bankrupt firms and other nonbankrupt industry firms. We estimate the regressions using a logistic-limited dependent variable model. The dependent variable equals 1 if a plant is closed in that year. The change in industry shipments and capacity utilization are yearly at the 4-digit SIC code level. Weighted industry research and development (R&D) is from COMPUSTAT firm-level data and represents R&D and advertising expense divided firm sales aggregated up to the 3-digit SIC code. Total Factor Productivity (TFP) is calculated using a translog production function. Operating cash flow is the value of shipments less labor, materials and energy costs divided by value of shipments. Relative plant scale is the plant's asset size divided by the average assets for plants in each industry. Capacity utilization is at the 4-digit SIC code and is from the Bureau of the Census. Data are yearly from 1977 to 1990. (*p*-values are in parentheses.)

	Dependent Variable:			Test for Significant Difference: Quartile 4 – Quartile 1 (<i>p</i> value)
	Plant Closing			
	Logit A: Full Sample	Logit B: Quartile 1	Logit C: Quartile 4	
Industry-level variables				
Change in industry shipments	–1.581 (0.000) ^a	–1.304 (0.002) ^a	–2.135 (0.000) ^a	(0.018) ^b
Weighted industry R&D	–7.360 (0.000) ^a	–3.286 (0.014) ^b	–2.904 (0.038) ^b	(0.015) ^b
Capacity utilization	–0.006 (0.000) ^a	–0.007 (0.006) ^a	0.001 (0.214)	(0.396)
Standard deviation of industry shipments	3.783 (0.000) ^a	3.256 (0.000) ^a	12.00 (0.000) ^a	(0.000) ^a
Plant- and firm-level variables				
Number of plants owned by firm	0.006 (0.000) ^a	0.005 (0.000) ^a	0.007 (0.000) ^a	(0.556)
Log of firm shipments	–0.048 (0.000) ^a	–0.066 (0.000) ^a	–0.027 (0.160)	(0.429)
Total factor productivity (TFP) (Lagged one year)	–0.173 (0.000) ^a	–0.162 (0.000) ^a	–0.279 (0.000) ^a	(0.000) ^a
Plant-level operating cash flow (Lagged one year)	–0.147 (0.002) ^a	–0.158 (0.014) ^b	–0.097 (0.296)	(0.226)
Relative plant scale	–0.276 (0.000) ^a	–0.286 (0.000) ^a	–0.463 (0.000) ^a	(0.000) ^a
Plant age	–0.022 (0.000) ^a	–0.026 (0.000) ^a	–0.030 (0.000) ^a	(0.261)
Bankruptcy variables				
Before Chapter 11 dummy variable (= 1 for years before firm is in Ch. 11)	0.681 (0.000) ^a	0.692 (0.000) ^a	0.944 (0.000) ^a	(0.663)
In Chapter 11 dummy variable (= 1 while firm is in Ch. 11)	–0.123 (0.916)	–2.019 (0.384)	–0.473 (0.889)	(0.899)
After Chapter 11 dummy variable (= 1 after firm emerges from Ch. 11)	0.382 (0.007) ^a	–0.136 (0.787)	0.529 (0.002) ^a	(0.056) ^b
In Chapter 11*TFP	–0.543 (0.000) ^a	–0.394 (0.000) ^a	–0.808 (0.000) ^a	

Table VIII – Plant Closing Decisions, Part 1

Read: Table VIII estimates plant closure decisions. Industry shipment growth and industry R&D are negatively related to closure. Plant productivity, plant cash flow, relative scale, and plant age are also negatively related to closure.

Interpretation: Chapter 11 status itself does not imply a broad failure to close plants. The in-Chapter 11 dummy is not significant in the full sample. However, the interactions show that during Chapter 11, closure decisions become more sensitive to changes in industry shipments and industry R&D. This is consistent with bankrupt plants being marginal capacity.

Takeaway: Plant closures look economically disciplined. Low productivity, low cash flow, and poor industry conditions predict closures. This weakens the view that Chapter 11 generally keeps inefficient plants alive.

Table VIII—Continued

	Dependent Variable: Plant Closing			Test for Significant Difference: Quartile 4 – Quartile 1 (<i>p</i> value)
	Logit A: Full Sample	Logit B: Quartile 1	Logit C: Quartile 4	
Bankruptcy variables				
In Chapter 11*cash flow	–0.804 (0.189)	–0.293 (0.751)	–0.418 (0.351)	(0.966)
In Chapter 11*industry R&D	–23.915 (0.015) ^b	–70.289 (0.008) ^a	–41.85 (0.069) ^c	(0.008) ^a
In Chapter 11*change in shipments	–3.072 (0.035) ^b	–9.213 (0.054) ^c	–10.51 (0.025) ^b	(0.393)
In Chapter 11*capacity utilization	0.007 (0.216)	0.032 (0.262)	0.031 (0.452)	(0.337)
Total plant years	371,373	60,477	85,148	
χ^2 statistic	1299.79	145.33	463.81	
Significance level (<i>p</i> -value)	<1%	<1%	<1%	

Table VIII – Plant Closing Decisions, Part 2

7.7 Table IX – Plant Closures: Estimated Probabilities; Table X – Transfers to Chapter 7 Liquidation, Part 1

Table IX
Plant Closures: Estimated Probabilities

Estimated probabilities of a plant closure by bankrupt and nonbankrupt firms at the 10th, 25th, 50th, and 90th percentiles of change in industry shipments. We compute estimated probabilities using the coefficients from the logit regressions from Table VIII using logit regression A for the full sample and logit regressions C and D for the quartiles. We compute these probabilities holding all other variables besides change in industry shipments and its interaction terms at the sample means of the nonbankrupt and bankrupt firms. For the year $t - 1$, nonbankrupt firms and the bankrupt firms' probabilities are computed with the before-bankruptcy indicator variable equal to 0 and 1 respectively. For the year $t + 1$ for the bankrupt firm, we include the during-bankruptcy indicator variable and the interactions with the industry and productivity variables. For quartiles, the percentiles refer to that specific subsample. Thus the 50th percentile for quartile 1 is the 12.5th percentile of the overall sample.

		Probability of Plant Closure Varying Change in Industry Shipments			
		10th Percentile	25th Percentile	50th Percentile	90th Percentile
Bankrupt firm: All quartiles (data from bankrupt sample)	Year $t - 1$	3.65%	3.38%	3.23%	2.86%
	Year $t + 1$	1.92%	1.62%	1.43%	1.22%
Quartile 1	Year $t + 1$	12.69%	9.81%	9.50%	6.29%
Quartile 4	Year $t + 1$	0.46%	0.30%	0.25%	0.15%
Bankrupt firm: All quartiles (data from nonbankrupt sample)	Year $t - 1$	3.23%	3.04%	2.89%	2.45%
	Year $t + 1$	1.66%	1.33%	1.19%	1.04%
Quartile 1	Year $t + 1$	11.76%	8.55%	8.60%	6.09%
Quartile 4	Year $t + 1$	0.39%	0.29%	0.19%	0.05%
Nonbankrupt firm: (data from nonbankrupt sample)	Year $t - 1$	1.65%	1.55%	1.48%	1.25%
	Year $t + 1$	1.57%	1.45%	1.40%	1.23%
Nonbankrupt firm: (data from bankrupt sample)	Year $t - 1$	1.87%	1.73%	1.65%	1.45%
	Year $t + 1$	1.83%	1.73%	1.66%	1.47%

Table IX – Plant Closures: Estimated Probabilities

Table X
Transfers to Chapter 7 Liquidation

This table examines the productivity of firms' plants that transferred into Chapter 7 liquidation from Chapter 11 compared to the remaining Chapter 11 firms that did not enter Chapter 7. Numbers are relative total factor productivity (TFP) for plants less the industry-average TFP for each year. TFPs are standardized by dividing by the standard deviation of industry TFP at the four-digit SIC code level. Quartiles are determined by using the change in the real value of shipments at the three-digit SIC code from the NBER. Productivity (TFP) numbers are presented combining the lowest two and highest two quartiles to obtain sufficient numbers of plants to allow disclosure under government criteria. — means that the cell cannot be disclosed because there are too few plants in this relative year. The Chapter 11 firms in columns 1 and 3 are the subset of Chapter 11 firms that do not transfer to Chapter 7. Columns 2 and 4 represent the years in Chapter 7 after transferring from Chapter 11, for the lowest two and highest two quartiles, respectively. We assign firms to Chapter 7 based on the year-end status; thus some firms transfer to Chapter 7 in the same year they declare Chapter 11. (t -statistics for significant differences from zero are in parentheses.)

Growth in Shipments								
Lowest Two Quartiles					Highest Two Quartiles			
	Chapter 11 Firms	<i>n</i>	Ch. 11 Firms transferred to Chapter 7	<i>n</i>	Chapter 11 Firms	<i>n</i>	Ch. 11 Firms transferred to Chapter 7	<i>n</i>
Panel A: Years before Chapter 11								
Year - 2	-0.031 (-0.82)	636	-0.219 (-2.81) ^a	146	0.046 (0.88)	330	-0.198 (-1.75) ^c	76
Year - 1	-0.036 (-1.04)	607	-0.098 (-1.20)	109	-0.019 (-0.33)	295	-0.189 (-1.38)	64
Panel B: After Declaration of Chapter 11								
Year 0	-0.100 (-2.35) ^c	514	-0.070 (-0.40)	24	-0.046 (-0.77)	299	-0.347 (-2.58) ^a	36
<i>t</i> -stat for average	(-2.35) ^c		(-0.40)		(-0.77)		(-2.58) ^a	
<i>t</i> -stat for -1 to 0	(-1.15)		(0.15)		(-0.32)		(-0.82)	
Year + 1	0.046 (0.89)	412	-0.105 (-0.41)	13	0.016 (0.19)	209	—	
<i>t</i> -stat for average	(0.89)		(-0.41)		(0.19)			
<i>t</i> -stat for -1 to +1	(1.32)		(-0.03)		(0.37)			
Year + 2	0.026 (0.46)	208	—		-0.073 (-0.78)	129	—	
<i>t</i> -stat for average	(0.46)				(-0.78)			
<i>t</i> -stat for -1 to +2	(0.93)				(-0.58)			
Year + 3	0.048 (0.62)	133	—		-0.336 (-2.49) ^a	59	—	
<i>t</i> -stat for average	(0.62)				(-2.49) ^a			
<i>t</i> -stat for -1 to +3	(0.99)				(-2.96) ^a			
Year + 4	0.072 (0.76)	85	—		-0.177 (-1.32)	49	—	
<i>t</i> -stat for average	(0.76)				(-1.32)			
<i>t</i> -stat for -1 to +4	(1.07)				(-1.28)			

Table X – Transfers to Chapter 7 Liquidation, Part 1

Read: Table IX gives the economic size of closure effects. At the 50th percentile, the estimated closure probability for bankrupt firms in quartile 1 is about 9.5%, compared with about 0.25% in quartile 4. This is a very large difference across industry states.

Interpretation: Table X separates firms that remain in Chapter 11 from those that transfer to Chapter 7. Before and during bankruptcy, plants of firms that later transfer to Chapter 7 are noticeably less productive, especially in the higher-growth quartiles.

Takeaway: Chapter 11 is associated with capacity exit in declining industries, while Chapter 7 transfers identify genuinely weak firms and plants. The bankruptcy system appears to sort severe cases toward liquidation.

7.8 Table X – Transfers to Chapter 7 Liquidation, Part 2

Table X—Continued

	Growth in Shipments							
	Lowest Two Quartiles				Highest Two Quartiles			
	Chapter 11 Firms	<i>n</i>	Ch. 11 Firms transferred to Chapter 7	<i>n</i>	Chapter 11 Firms	<i>n</i>	Ch. 11 Firms transferred to Chapter 7	<i>n</i>
Panel C: After Leaving Chapter 11, Emerged Firms versus Firms Transferred to Chapter 7								
Year +1	−0.087	260	−0.274	80	−0.207	112	−0.131	38
<i>t</i> -stat for average	(−1.41)		(−2.89) ^a		(−2.32) ^b		(−0.97)	
<i>t</i> -stat for −1 to +1	(−0.71)		(−1.40)		(−1.78) ^c		(0.31)	
Year +2	−0.104	136	−0.197	69	−0.073	77	−0.409	17
<i>t</i> -stat for average	(−1.21)		(−1.65) ^c		(−0.86)		(−1.99) ^c	
<i>t</i> -stat for −1 to +2	(−0.73)		(−0.68)		(−0.53)		(−0.64)	
Year +3	−0.126	75	−0.389	69	−0.226	49	−0.583	15
<i>t</i> -stat for average	(−0.92)		(−3.50) ^a		(−1.51)		(−2.67) ^a	

Table X – Transfers to Chapter 7 Liquidation, Part 2

Read: The continuation of Table X shows post-Chapter 11 outcomes for emerged firms versus firms transferred to Chapter 7. Plants of firms transferred to Chapter 7 remain much less productive than plants of firms that emerge.

Interpretation: This evidence supports the view that liquidation is not random. The bankruptcy process identifies lower-productivity plants and firms for Chapter 7 liquidation.

Takeaway: Including Chapter 7 converters avoids survivorship bias and strengthens the paper’s conclusion that bankruptcy outcomes reflect economic efficiency and industry conditions more than uniform Chapter 11 distortions.

8 How to read the paper’s causal claims

The paper is not a randomized experiment. The empirical claims rely on detailed conditioning and comparison:

- **Within-industry productivity benchmarking** compares plants to industry peers rather than to all manufacturing plants.
- **Industry-growth quartiles** isolate different demand environments.
- **Event-time analysis** separates pre-bankruptcy selection from within-bankruptcy changes.
- **Plant-level tracking** avoids the survivor-firm bias that would arise if sold and closed plants disappeared from the data.
- **Logit regressions and estimated probabilities** quantify whether Chapter 11 status has incremental explanatory power after controlling for industry, plant, and firm characteristics.

Takeaway: The most credible claim is not that bankruptcy has zero costs in all states. The stronger and better-supported claim is that observed Chapter 11 outcomes are explained primarily by economic fundamentals, especially industry demand and plant productivity.

9 Key lessons for future research

1. **Bankruptcy research must condition on industry state.** Pooling high-growth and declining industries can hide important mechanisms.
2. **Plant-level data are essential.** Firm-level data cannot distinguish a productivity decline from selective asset sales and closures.
3. **Real costs of bankruptcy are endogenous.** Firms may avoid Chapter 11 when the indirect costs would be high. Observed Chapter 11 cases therefore need not reveal costs for all distressed firms.
4. **Asset specificity and R&D matter.** The R&D results suggest that redeployment costs and technological fit influence whether plants are sold or closed.
5. **Chapter 11 can be an efficient capacity-exit mechanism.** In declining industries, it seems to help remove capacity rather than protect inefficient plants.

10 Short summary

- The paper studies 302 Chapter 11 manufacturing firms and their plants using U.S. Census plant-level data.
- It asks whether Chapter 11 imposes indirect costs through productivity declines or inefficient asset reallocation.
- The main result is that industry conditions dominate bankruptcy status in explaining productivity, asset sales, and plant closures.
- In declining industries, bankrupt plants do not significantly underperform or deteriorate relative to industry peers; Chapter 11 mainly appears to facilitate capacity exit.
- In high-growth industries, bankrupt plants are more likely to be inefficient, and sold plants can improve under new owners.
- The paper concludes that realized real costs of Chapter 11 are limited for firms that choose to file, but it leaves open that firms outside Chapter 11 may avoid filing because expected indirect costs would be high.